Git & GitHub

Comprehensive student notes | Fundamentals, guided practice, collaboration, and troubleshooting

These notes accompany the live workshop. Examples use PowerShell on Windows. Git commands themselves are largely the same on macOS and Linux; file-creation and navigation commands shown here use PowerShell syntax. Type commands in a terminal, not in a browser or a Markdown file.

What you will be able to do	Evidence of learning
Track a local project	Create a repository, inspect status, stage changes, and make commits.
Publish and collaborate	Connect a GitHub remote, push a branch, open a pull request, and pull a merge.
Recover safely	Inspect differences, unstage a file, discard a known edit, and resolve a simple conflict.

Suggested study route

Read sections 1-4 before the hands-on session. Follow sections 5-9 at the terminal. Use sections 10-14 as a reference and attempt the exercises and quiz afterward.

1. What problem does version control solve?

A project changes over time. A change may improve the project, break it, or need review. Git records selected snapshots called commits, allowing a team to examine what changed, who recorded it, and which earlier state a later state builds upon. It also supports separate lines of work called branches.

Git is software that works on your computer. GitHub is a hosting and collaboration service for Git repositories. You can use Git without GitHub and can browse GitHub without installing Git. A repository is a project directory that Git tracks along with its history.

Action	Where it happens	What it means
Save a file	Working directory	Writes the current file contents to disk.
git add	Staging area (index)	Selects a snapshot of file contents for the next commit.
git commit	Local repository	Records the staged snapshot as a local commit.
git push	GitHub remote	Sends local commits to the remote repository.
git pull	Local repository	Fetches remote changes and integrates them into the current branch.

Key point. Saving, committing, and pushing are three different actions. A commit exists locally even before you push it.

2. The four places to keep in mind

Place	Meaning	Useful command
Working directory	The files you can open and edit.	git status; git diff
Staging area	The proposed content of the next commit.	git add; git diff --cached
Local repository	Commits and branches stored in the .git directory.	git log; git branch
Remote repository	The GitHub copy associated with a remote such as origin.	git remote -v; git push; git fetch

Typical flow: edit and save a file; inspect its changes; stage the intended version; commit it with a descriptive message; push that commit if you want it on GitHub. Git can track the same file through all these places at once. A file can even have staged changes and newer unstaged edits at the same time.

Terms that appear in Git output

Untracked: Git sees a new file but has never staged it. Modified: a tracked file differs from a saved version. Staged: content is selected for the next commit. Clean: no uncommitted changes are detected. HEAD: your currently checked-out commit, usually through the current branch. origin: a conventional nickname for a remote URL, not a special GitHub account.

3. Setup and first checks

Install Git, create a GitHub account, and open PowerShell. Verify Git and set the name and email that will be written into future commits. These settings do not sign you into GitHub.

```
git --version
$GitName = Read-Host "Enter your name for Git commits"
$GitEmail = Read-Host "Enter your email for Git commits"
git config --global user.name "$GitName"
git config --global user.email "$GitEmail"
git config --global --get user.name
git config --global --get user.email
```

Use an email verified in your GitHub account, or your GitHub no-reply email, if you want GitHub to attribute commits to your profile. GitHub authentication for pushing is a separate sign-in step. Never paste a password or token into a repository file.

Global versus local configuration

--global sets a default across repositories for your computer account. To use a different identity in one repository, run git config user.name "Name" and git config user.email "address" from inside that repository, without --global. Local configuration takes precedence there.

4. Make a repository and your first commit

Start in a suitable parent folder. These commands assume Git-Hub-Workshop does not already exist. If it does, choose another repository name and use it consistently for the GitHub step.

```
New-Item -ItemType Directory -Name Git-Hub-Workshop
Set-Location Git-Hub-Workshop
git init -b main
git status
```

Git creates the hidden .git directory. Do not manually edit its internals. The branch main starts without commits. Older Git versions that do not support git init -b main can use git init followed by git branch -M main.

Create a README

A README.md is the front page of a GitHub repository. The content below is an example. The PowerShell here-string starts with @' and closes with '@ at the start of the last line as shown. Replace the name with your own if desired.

```
@'
# Student Project Showcase

A collection of projects and ideas I am exploring.

## About me

Name: Your Name
Interest: AI systems and web development

## Project ideas

- NexusAgent Studio: A developer workspace for AI-assisted projects.
'@ | Set-Content -Encoding utf8 README.md
```

```
git status
git add README.md
git status
git diff --cached
git commit -m "Add initial project README"
git log --oneline
git status
```

The first status shows an untracked file. The second shows it staged. The commit ID in git log is unique to the content and history; yours will differ from a neighbor's. After the commit, status should show a clean working tree. If a command opens a pager displaying (END), press q to close it.

5. Edit a file and read a diff

A diff shows line-level changes. A line starting with + was added; one starting with - was removed. Headers identify the file and location. Before staging, git diff compares the working directory to the staging area. After staging, git diff --cached compares the staging area to the last commit.

```
Add-Content README.md "- StudySync AI: A study group collaboration app."
git status --short
git diff
git add README.md
git diff --cached
git commit -m "Add StudySync project idea"
git log --oneline
```

In short status, ?? means untracked; M in the left column means staged modification; M in the right column means unstaged modification. A newly added staged file appears as A on the left. Use full git status if the short form is unclear.

Key point. git add captures the file contents at that moment. Edit the file again and you must git add it again for the later edit to be included in the next commit.

6. Ignore local files and protect secrets

A .gitignore file lists patterns for untracked files that should normally be excluded. Typical examples include generated output, local notes, and environment files. It does not remove a file that was already tracked, and it cannot erase a secret from old commits.

```
@'
notes.local.txt
*.env
'@ | Set-Content -Encoding ascii .gitignore
Set-Content -Encoding ascii notes.local.txt "Personal notes for my computer only."
git status --short
git check-ignore notes.local.txt
git add .gitignore
git commit -m "Ignore local notes and environment files"
```

notes.local.txt is absent from status because it matches a rule. The .gitignore file itself is tracked after the commit. Review git status and git diff --cached before any commit. If a credential is accidentally published, revoke or rotate it promptly; simply deleting it in a new commit does not make the old value disappear from history.

7. Put the project on GitHub

In GitHub, create an empty repository called Git-Hub-Workshop. Do not initialize it with a README, .gitignore, or license because the local project already has its own initial commits. Choose public or private visibility intentionally. Then return to PowerShell:

```
$GitHubUser = Read-Host "Enter your GitHub username"
git remote add origin "https://github.com/$GitHubUser/Git-Hub-Workshop.git"
git remote -v
git push -u origin main
```

The first push can open a browser sign-in or credential helper prompt. -u sets the upstream relationship: local main tracks origin/main. Future pushes from the same branch can often use git push. Refresh the GitHub page to confirm the README and commits appear. A remote URL must match your actual account and repository name.

Clone versus init

Use git init when starting a new local project. Use git clone URL when a remote project already exists and you want a local working copy with a remote preconfigured. Cloning the repository above elsewhere would use the HTTPS URL shown by GitHub; do not clone inside an already initialized copy of the same project.

8. Branches, pull requests, and reviews

A branch is a named pointer to a sequence of commits. A feature branch allows you to develop a change separately from main. A pull request (PR) is a GitHub proposal to merge one branch into another; reviewers can inspect the diff and leave comments. Opening a PR does not by itself merge the change.

```
git switch -c feature/add-project-proposal
git branch --show-current
```

```
@'
# NexusAgent Studio project proposal

## Problem

Developers switch between tools while building projects.

## Proposed solution

Bring planning, coding, review, and tests into one workspace.

## First milestone

Build a small project dashboard and document the workflow.
'@ | Set-Content -Encoding utf8 project.md
git add project.md
git commit -m "Add project proposal"
git push -u origin feature/add-project-proposal
```

On GitHub, choose Compare & pull request, with base: main and compare: feature/add-project-proposal. Review Files changed, add a clear title and description, and create the PR. Reviewers can comment; the author can make another commit and push to the same branch, updating the PR. Merge it on GitHub when ready and permitted.

9. Bring the merged work back to your computer

```
git switch main
git status
git pull --ff-only origin main
git log --oneline --graph --decorate -6
Get-Content project.md
```

The remote merge does not automatically update your local main. git pull fetches and integrates the remote history. --ff-only accepts only a fast-forward update and stops if local and remote main have diverged; it avoids an unexpected merge commit. The exact graph depends on the PR merge method.

Command	When to use it
git fetch origin	Update remote-tracking information without merging into your current branch.
git pull --ff-only origin main	Update a local main that is behind origin/main and has not diverged.
git push	Publish local commits to the configured upstream branch.

10. Merge conflicts: how to handle one

A conflict can occur when Git cannot automatically combine changes, often because two branches edited the same lines. Do not panic or delete conflict markers without reading both sides. The safe process is to inspect status, open each conflicted file, decide what content belongs in the final file, save it, stage it, and complete the merge.

```
git status
git diff
# Open the conflicted file in your editor and resolve its content.
git add README.md
git status
git commit
```

The # line above is an instruction comment, not an action needed in Git. In the file, markers such as <<<<<<<, =======, and >>>>>>> delimit competing content. Remove the markers after choosing or combining the intended lines. git commit completes a normal merge after resolution. If you want to abandon an in-progress merge before committing, git merge --abort generally restores the pre-merge state; inspect git status first.

A git pull --ff-only refusal is different: Git has not started a merge and there are no conflict markers to resolve. First inspect git status and git log --oneline --graph --all -10, then decide with your team how to integrate the divergent histories.

11. Undo and recovery, with scope clearly stated

Situation	Command	What it does
Staged the wrong file; keep edits	git restore --staged README.md	Removes that file from staging; keeps working edits.
Discard known unstaged edits	git restore README.md	Replaces working file with its staged version; edit loss is possible.
View an earlier committed file	git show HEAD~1:README.md	Prints the file as stored one commit before HEAD.
Undo a shared commit with a new commit	git revert COMMIT_ID	Records an inverse change; may need conflict resolution.

Inspect git status and git diff first. Do not run git restore README.md if you still need the current unstaged work. Do not use git reset --hard or force-push as an ordinary fix in a shared project. Git cannot recover an edit that was never saved or committed in every circumstance.

12. Troubleshooting common messages

Symptom	Likely cause	Next check
fatal: not a git repository	Terminal is outside the project.	Get-Location; Set-Location Git-Hub-Workshop
nothing to commit, working tree clean	Nothing new is staged or saved.	git status; inspect the file and branch.
pathspec did not match any files	File does not exist at that path or name.	Get-ChildItem; verify spelling and current directory.
Author identity unknown	Name or email was not configured.	git config --global --get user.email
remote origin already exists	An origin remote is already registered.	git remote -v; use the existing correct remote.
Repository not found or permission denied	Wrong URL, missing access, or sign-in problem.	git remote -v; verify GitHub repository and account.
Push rejected: non-fast-forward	Remote contains changes absent locally.	git fetch origin; inspect git log --all --graph.
(END) in the terminal	Git opened a pager.	Press q.

Read the full error before trying another command. Ask a partner or instructor to check your current directory, branch, status, and remote URL; those four facts solve many beginner issues.

13. Command reference

Command	Meaning
git status	Summarize branch and working/staged changes.
git diff	Inspect unstaged edits.
git diff --cached	Inspect staged content.
git add FILE	Stage the current contents of FILE.
git commit -m "MESSAGE"	Create a local commit of staged content.
git log --oneline	Read condensed local history.
git switch -c BRANCH	Create and switch to a new branch.
git switch BRANCH	Move to an existing local branch.
git branch --show-current	Print the active branch name.
git remote -v	List remote nicknames and URLs.
git push -u origin BRANCH	Publish BRANCH and record its upstream.
git pull --ff-only origin main	Bring remote main into local main when fast-forward is possible.

The uppercase words FILE, MESSAGE, and BRANCH in the reference are placeholders to replace with real values; they are not literal workshop commands. Commands inside the guided practice use actual filenames and branch names.

14. Practice tasks and self-check

Independent challenge

From an up-to-date main branch, create feature/learning-goals. Add learning-goals.md with three concrete goals. Check status and diff, stage only that file, commit it, push the branch, and open a PR into main. In the PR description, explain what was added and how you checked it. Leave the PR open for peer review if your instructor requests that.

```
git switch main
git pull --ff-only origin main
git switch -c feature/learning-goals
# Create and save learning-goals.md in your editor.
git status
git add learning-goals.md
git diff --cached
git commit -m "Add semester learning goals"
git push -u origin feature/learning-goals
```

Questions

1. Does saving a file create a commit? 2. What does git add capture? 3. Which diff shows what will be committed? 4. Does git commit upload to GitHub? 5. What does origin name? 6. What are the base and compare branches in a PR? 7. Does .gitignore remove an already committed secret? 8. What should you check before discarding an edit?

Answers

1. No. 2. The selected file contents at that moment. 3. git diff --cached. 4. No; git push uploads commits. 5. A remote nickname, conventionally the main GitHub repository. 6. Base is the destination; compare is the proposed source. 7. No; rotate the secret and address history separately. 8. git status and git diff, then confirm that the change is no longer needed.

15. Glossary and teamwork checklist

Term	Plain-language meaning
Repository	The project and its Git history.
Commit	A recorded local snapshot with a message and parent history.
Branch	A named pointer to a line of development.
Remote-tracking branch	Your local record of a remote branch, such as origin/main.
Upstream	The branch used by default for push/pull on a local branch.
Pull request	A proposal and discussion for merging changes on GitHub.
Merge conflict	A change Git cannot combine without a decision.

Before asking for review

Check the active branch; inspect git status and your staged diff; make a focused commit with a meaningful message; push the feature branch; confirm the PR base and compare directions; write what changed and how you checked it. After review, respond to comments and push updates to the same branch. After merging, switch to main and pull the merged history before starting a new branch.

Further reading

Official references: Git documentation; Pro Git book; GitHub Hello World; GitHub pull request documentation. These are useful when a command behaves differently in your Git version or repository settings.